SUPERXfer

Data Transfer Script for SUPER

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P-207 Solar Generation

Supported data historian:	PI
Supported File Transfer Protocol:	SFTP
Python-PI Connect Library:	PIConnect

These instructions are to help users setup and run the EPRI SUPERXfer code. Steps 1-3 focus on installing Python and VSCode. The environment uses Python 3.12.x so the user may still need to download a 3.12 version of Python even if they already have a different version of Python installed on their computer. If a user has Python 3.12 and an IDE already installed, they can skip to step 4.

Step 4 focuses on setting up the environment using the requirements.txt file.

Step 5 gives instructions on setting up the user-specific inputs, including the channel list and SFTP credentials.

The Additional help section at the end of the document provides instructions on opening a specific project in VSCode from CMD and on generating a channel list using the PIDataLink plugin with Microsoft Excel.

These instructions are primarily written for Windows, but equivalent commands for other operating systems can be found online. If you need assistance with setting up the script, feel free to reach out to EPRI.

Step 1: To check if you have a Python 3.12 version installed, open the Command Prompt and Run:

py -3.12 --version
Python 3.12.0

- This will either output a version of Python 3.12 that you already have installed or give you an error saying something like: "No suitable Python runtime found".

Step 2: Python Install

1. Install Python:
   (a) Download the "Windows x86-64 executable installer" from here. For this project, select a version starting with 3.12.
   (b) Run the installer and choose "Customize installation" when prompted.

(c) Recommended optional features:
   - Check "Documentation"
   - Check "pip"
   - Check "tcl/tk and IDLE"
   - Check "Python test suite"
   - Check "py launcher" and uncheck "for all users"

(d) Recommended advanced options:
   - Uncheck "Install Python for all users"
   - Check "Associate files with Python"
   - Check "Create shortcuts for installed applications"
   - Check "Add Python to environment variables"
   - Check "Precompile standard library"
   - Uncheck "Download debugging symbols"
   - Uncheck "Download debug binaries"

(e) After installation you may be prompted with this option: "Disable path length limit." You can skip that prompt.

(f) Open Command Prompt and confirm py launcher works for your specific Python version. Run:

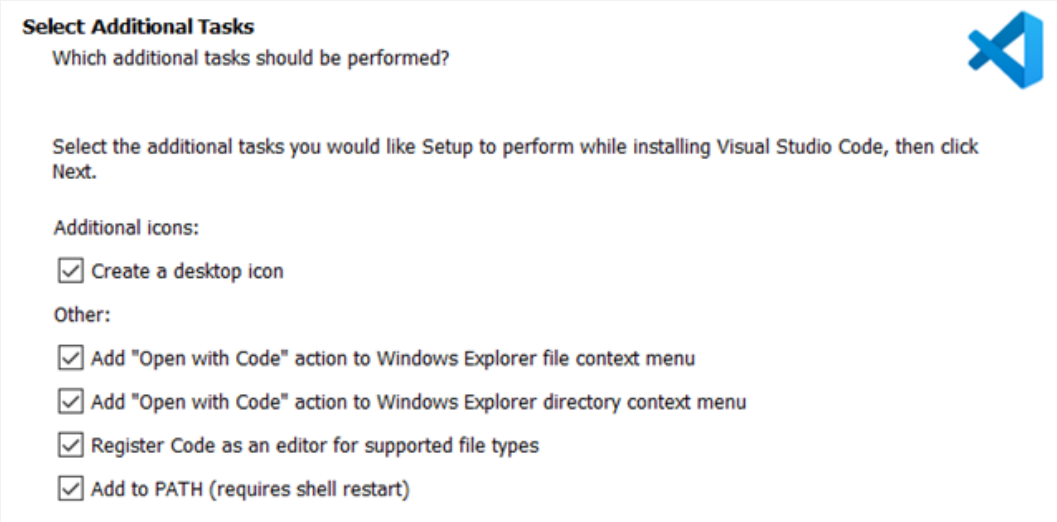
```
py -3.12 --version
```

This should output the version of Python you just downloaded and installed.

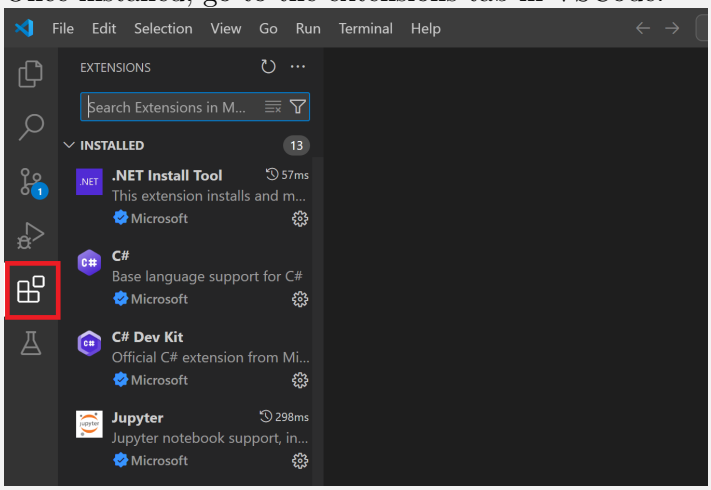
If you already have a preferred IDE other than VSCode, please skip the VSCode-related instructions and use the IDE you are most comfortable with.

Step 3: VSCode Install

1. Go to VSCode Installer the vs code installer for your machine
2. Check all the boxes when prompted (if adding VSCode to PATH causes an admin error, restart the install and deselect that option):



3. Once installed, go to the extensions tab in VSCode:



4. Extensions to install:
   (a) Python (by Microsoft)
   (b) Pylance (by Microsoft)

Step 4: Environment Setup

1. Open the Command Prompt (Windows) or Terminal (Mac)
2. Navigate to the directory that already contains or will contain the target project code.
3. Create a virtual environment (this command may take a couple of minutes). Example:
   (a) **Windows:** py -m venv env-superxfer
   (b) **Mac/Linux:** python3 -m venv env-superxfer
4. Active the virtual environment. Example:

(a) **Windows:** env-superxfer\Scripts\activate.bat
(remove .bat if the command doesn't work. You should see the environment's name in parenthesis in the command prompt or terminal)

(b) **Mac/Linux:** source env-superxfer/bin/activate

5. Install packages from the requirements.txt file.

py -m pip install -r requirements.txt

Step 5: Run the Script

1. Create a channel list spreadsheet according to the example in the `Channel_List` directory. Save the new list in the same directory and name it something like `CompanyABC_PV_channel_list.xlsx`. The additional help section below shows how to populate a channel list spreadsheet using PIDataLink.

2. Create an SFTP account after receiving email instructions from EPRI. (Contact DFregosi@epri.com and/or KBuch@epri.com)

3. EPRI will send an SSH key associated with the SFTP account. Save the SSH key in the `"SSH_KEYS"` folder.

4. Edit `constants.env` and complete the "Variables to update" Descriptions are given in the file for each variable..

5. **(Set up the virtual environment first and activate it before proceeding to this step.)** Execute the code to transfer data. Monitor the progress. Notify EPRI (DFregosi@epri.com and/or KBuch@epri.com) if the script encounters an error. Subsequent executions will resume data transfer starting from where the previous transfer ended.

Additional Help:
Here are some ways to open the a specific project in VSCode :

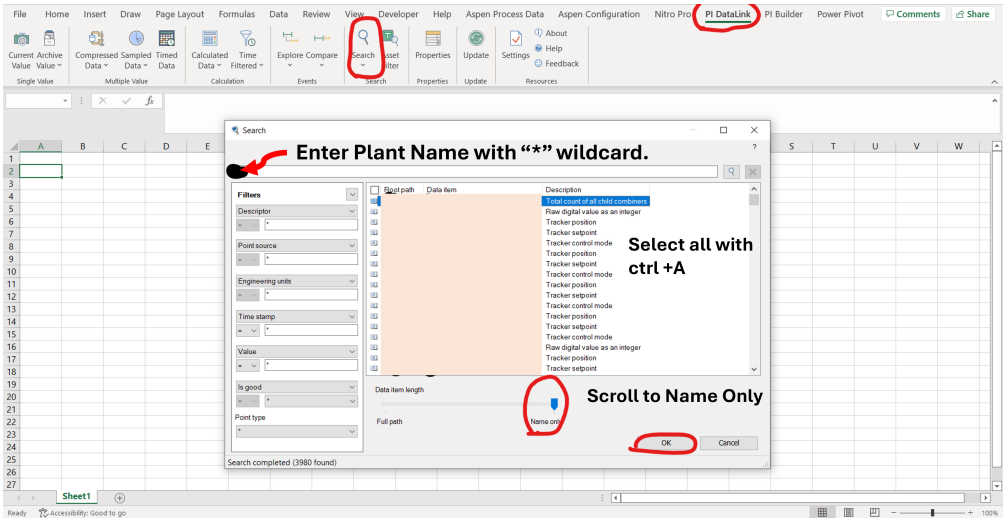
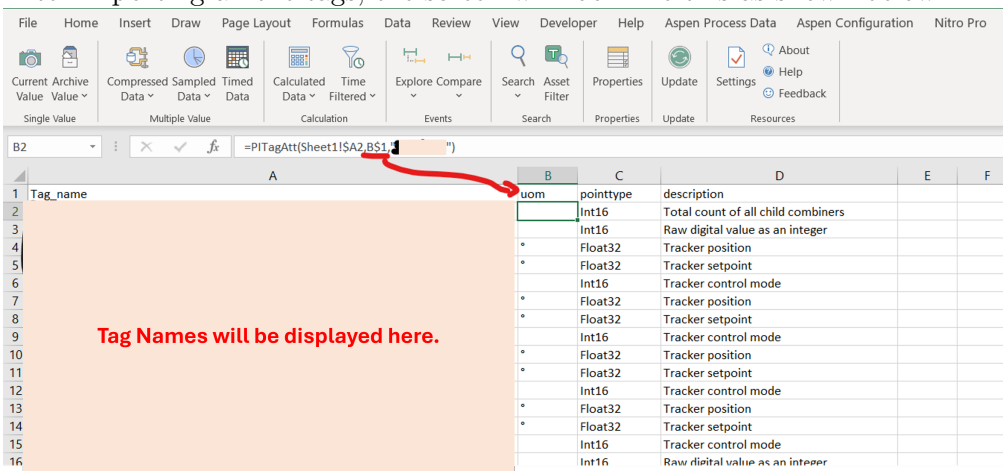
1. You can either use the command prompt and navigate to the folder you would like to work on code in using `"cd <folder_name>"` until you arrive at the target folder. Once there, type `"code ."` to open VSCode within that directory.

2. Another option is to use the File Explorer and navigate to the project folder, then right-click and select "Open with Code."

3. You can also open VSCode and then choose "Open Folder" in the File option and open the project folder that way.

Here is one of the ways to export the tags from the PI System using PIDataLink and Excel:

1. Open Microsoft Excel with `PIDdataLink` plug-in and follow the steps illustrated below:



2. After importing all the tags, the screen will look like this as shown below: